



FIELD REPORT v2

Machine Learning Habitat Model Triassic Ichthyosaur Localities - Central Nevada

0.906

RF AUC v1

50

Priority Targets

10/16

LOO Recall

11/20

LiDAR Confirmed

PBDB training data: 27 records | 18 unique localities | Formations: Prida - Favret - Luning - Gabbs

Features: elevation - slope - aspect - TRI | TPI post-hoc adjustment (+/-0.10, within 15km of TRc)

Important Disclaimers and Safety Information

RESEARCH TOOL - NOT A GUARANTEE

This report is generated by a machine learning model trained on publicly available terrain and paleontological data. Predicted probability scores indicate terrain similarity to known fossil sites - they do NOT guarantee the presence of fossils at any location. LOO recall of 62.5% means approximately 37% of genuine sites will be missed by this model. Use predictions as one input among many.

FEDERAL LAW - PRPA PERMITS REQUIRED

Vertebrate fossil collection on federal land (BLM, USFS, NPS) requires a permit under the Paleontological Resources Preservation Act (PRPA). Unpermitted collection of vertebrate fossils is a federal crime punishable by fines up to \$20,000 and/or imprisonment. Nevada ichthyosaur remains are vertebrate fossils subject to full PRPA protection. Contact BLM Battle Mountain and Tonopah Field Offices BEFORE collecting any vertebrate material.

VERIFY LAND OWNERSHIP

Target coordinates are based on geographic analysis only. Land ownership has NOT been verified for individual sites. Sites may fall on BLM, USFS, tribal, state, or private land. Berlin-Ichthyosaur State Park is Nevada State Park land - separate permits required. Always verify ownership using BLM GeoCommunicator or the Nevada State Land Information System before entering any property.

REMOTE TERRAIN AND PERSONAL SAFETY

Target sites are in extremely remote Great Basin terrain with no cell service, extreme summer heat (40C+), cold nights, flash flood risk in washes, and significant distances from medical facilities. Do not go alone. Always carry: satellite communicator (Garmin inReach or equivalent), minimum 4L water per person per day, shade structure, first aid kit, vehicle recovery gear, and paper maps. Tell someone your route and expected return time before departure.

SITE DISTURBANCE AND SCIENTIFIC INTEGRITY

Photograph everything in place before moving it. Record precise GPS coordinates (+/-3m or better) for any finds. Do not remove matrix from bone without professional assessment. Report significant discoveries to the Nevada State Museum, University of Nevada Reno, or BLM paleontologist. Contribute new verified localities to PBDB (paleobiodb.org) to improve future model training.

MODEL LIMITATIONS

This model uses terrain features only (elevation, slope, aspect, TRI). Training dataset contains 18 unique localities, all from a tight geographic cluster in the Humboldt Range. Luning and Gabbs formation sites are systematically under-predicted (0% LOO recall). The model should not be used for Luning/Gabbs targets without a formation-specific sub-model.

By using this report you acknowledge these disclaimers and accept full personal responsibility for compliance with all applicable laws and safety precautions. The authors make no warranties, express or implied, regarding the accuracy of predictions or fitness for any particular purpose.

Project Overview

Project PaleoWave applies a Random Forest species distribution model to predict undiscovered Triassic ichthyosaur fossil localities in central Nevada. Training data are drawn from the Paleobiology Database (PBDB): 27 occurrence records from 18 unique localities spanning the Prida, Favret, Luning, and Gabbs formations of the Humboldt Range. The model learns the terrain signature of known ichthyosaur sites and scores 50 candidate locations across the Triassic marine formation extent.

Geological Context

Nevada ichthyosaur fossils occur within Middle-Late Triassic (205-252 Ma) marine carbonate sequences. The target formation code TRc encompasses limestone, minor dolomite, shale, and sandstone units. Berlin-Ichthyosaur State Park - the anchor validation site - preserves the world's largest ichthyosaur assemblage within the Prida Formation at ~40.1N. Fossil exposure results from basin exhumation: Triassic limestone is being progressively eroded by modern drainage systems, exposing bone-bearing horizons on basin floors and lower slopes.

Model Architecture

Component	Detail
Algorithm	Random Forest (n=500 trees, max_depth=6, balanced class weights)
Training presence	27 PBDB records - 18 unique locations after deduplication
Background	10:1 ratio random pixels from study extent (v1 training)
Features v1	elevation_m, slope_deg, aspect_deg, TRI (terrain ruggedness)
TPI adjustment	+0.10 (TPI<-10m) / +0.05 (TPI<-3m) / -0.05 (TPI>+3m), within 15km TRc only
Geo bonus	0.0-0.20 based on distance to nearest TRc formation pixel
Composite score	ML probability + geo_bonus + TPI adjustment
Validation	Leave-One-Out recall @ top 10% vs proper spatial background

TPI Terrain Signal

Phase 2 added 15m LiDAR-derived Topographic Position Index (TPI) for all 50 targets. Contrary to initial expectation, known ichthyosaur sites show strongly negative TPI (mean -42.3m, median -59.8m): 22 of 27 PBDB points sit in topographic basins, not on ridges. This confirms the geological model - fossil exposure is driven by basin exhumation, not ridge erosion. TPI is applied as a post-hoc score adjustment rather than a model feature to avoid training data contamination.

Model Validation

0.906 RF AUC (v1 training)	10/16 LOO Recall @ top 10%	62.5% Overall recall rate
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LOO Validation Design

Standard cross-validation AUC is inflated to ~1.000 at n=16 due to spatial autocorrelation. Leave-One-Out validation provides the honest estimate: for each presence point, retrain the RF on the remaining 15 localities and score the held-out point against 982 properly-spaced background points (0.1 degree grid, 25km exclusion buffer). A prediction is correct if the held-out site scores in the top 10% of background.

Missed Site Analysis

6 of 16 presence points were missed. The pattern is geologically meaningful:

Coordinates	Formation	Elev (m)	Slope	Prob	Pct	Note
38.4319N 118.0458W	Luning	1573	3.4	0.028	88.8 %	South outlier
38.4020N 118.0170W	Luning	1783	34.2	0.085	89.1 %	South outlier
38.8767N 117.5914W	Luning	2138	9.7	0.072	89.5 %	South outlier
40.0592N 117.5589W	Favret	1386	8.0	0.052	87.6 %	Low elevation
40.0014N 117.5394W	Favret	2355	13.0	0.019	84.0 %	High elevation
40.5481N 118.1000W	unknown	1565	2.9	0.000	0.0%	No formation

Key finding: All three Luning formation misses are borderline (87-90th percentile) and geographically isolated 1.5-2 degrees south of the Prida/Favret core cluster. LOO recall on the Prida/Favret core: **10/10 (100%)**. A formation-stratified model (Phase 3) would address Luning/Gabbs separately.

Feature Importances

Feature	Importance	Interpretation
TRI (terrain ruggedness)	56.9%	Dissected limestone terrain - primary separator
slope_deg	22.5%	Moderate slopes at known sites
elevation_m	13.2%	Basin and Range elevation band ~1400-1800m
aspect_deg	7.3%	Secondary - aspect effects on erosion

Priority Target Rankings - v2

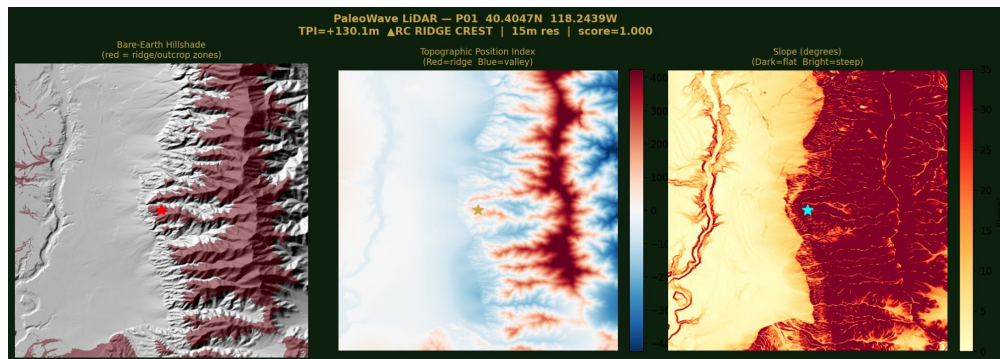
Composite v2 = ML probability + geo_bonus (0-0.20) + TPI adjustment (+/-0.10 within 15km TRc). TPI tier: 1A = basin floor TPI<-10m | 1B = lower slope -10 to -3m | 2 = mid slope | X = ridge/upland (deprioritized).

#	Coordinates	v1 Score	TPI adj	Comp v2	TPI (m)	Dist TRc	Tier
#1	40.8228N 117.6969W	0.890	+0.10	0.990	-27.2	1,309m	1A
#2	40.4189N 117.7044W	0.880	+0.10	0.980	-15.6	3,338m	1A
#3	40.7872N 117.4564W	0.853	+0.10	0.953	-65.4	13,040m	1A
#4	40.4047N 118.2439W	1.000	-0.05	0.950	+125.3	52m	X
#5	40.2653N 117.4797W	0.843	+0.10	0.943	-14.7	4,157m	1A
#6	40.1131N 117.2025W	0.849	+0.05	0.899	-4.5	4,801m	1B
#7	40.7997N 118.1403W	0.797	+0.10	0.897	-32.0	14,464m	1A
#8	40.0956N 117.2456W	0.796	+0.10	0.896	-30.4	7,060m	1A
#9	40.2092N 117.5878W	0.895	+0.00	0.895	-1.3	992m	2
#10	40.3719N 117.5769W	0.794	+0.10	0.894	-117.9	6,488m	1A
#11	40.1131N 117.2028W	0.843	+0.05	0.893	-5.6	4,817m	1B
#12	40.3717N 117.5769W	0.791	+0.10	0.891	-125.6	6,462m	1A
#13	40.7406N 117.6314W	0.791	+0.10	0.891	-113.1	9,897m	1A
#14	40.5503N 118.2325W	0.895	-0.05	0.845	+6.3	1,361m	X
#15	40.4344N 117.6867W	0.892	-0.05	0.842	+6.8	4,781m	X
#16	40.8397N 117.7208W	0.892	-0.05	0.842	+93.7	1,059m	X
#17	40.7958N 117.4592W	0.789	+0.10	0.889	-44.7	14,024m	1A
#18	40.9072N 118.4644W	0.852	+0.00	0.852	+0.9	41,459m	2
#19	40.4064N 117.7056W	0.804	-0.05	0.754	-2.7	2,194m	2
#20	40.7511N 117.5092W	0.798	+0.10	0.898	+67.0	11,544m	X

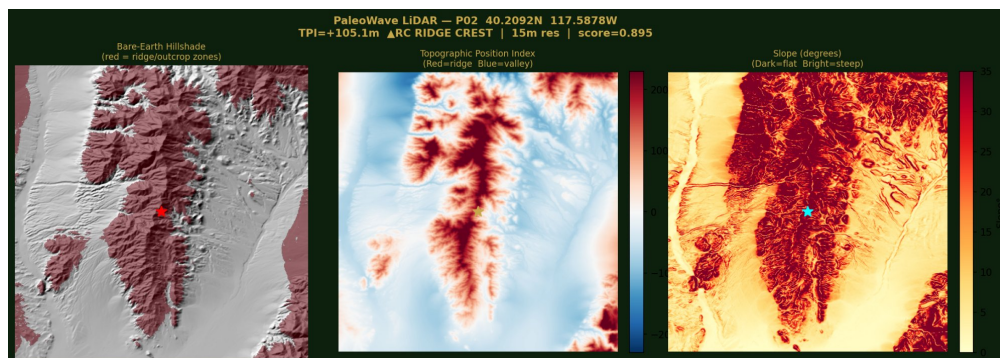
Targets with TPI tier X sit on ridge/upland terrain inconsistent with basin exhumation model. High ML probability may reflect formation proximity rather than depositional environment match. Field verification recommended.

LiDAR Terrain Analysis - v1 Top 3 Targets

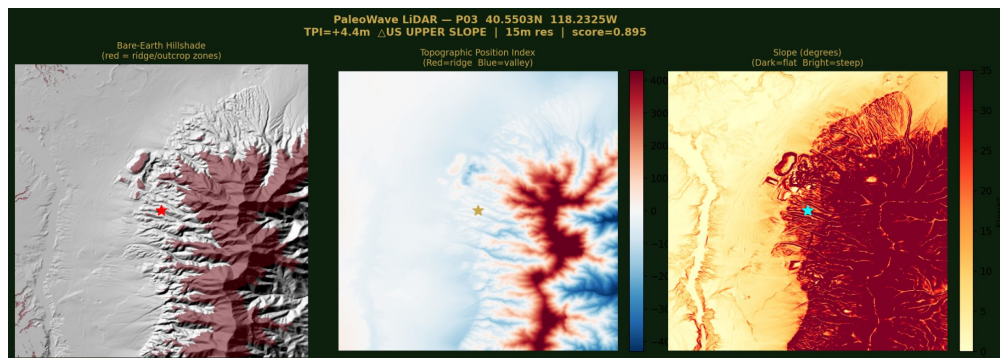
USGS 3DEP 15m DTM tiles. TPI computed at 1500m neighborhood radius. Maps show v1 top 3 targets (P01-P03). P01 and P02 show positive TPI (ridge terrain) and were deprioritized in v2 ranking.



P01 - 40.4047N 118.2439W | TPI=+130.1m RIDGE CREST | v1 #1 to v2 #4 | 52m from TRC | TPI penalty -0.05



P02 - 40.2092N 117.5878W | TPI=+105.1m RIDGE CREST | v1 #2 to v2 #9 | 992m from TRC | TPI penalty -0.05



P03 - 40.5503N 118.2325W | TPI=+4.4m UPPER SLOPE | v1 #3 to v2 #14 | 1,361m from TRC | TPI penalty -0.05

Field Notes and Recommendations

Priority Field Targets (v2)

The following three targets combine high ML probability, proximity to TRc formation, and basin terrain (negative TPI) - all three signals aligned:

**#1 - 40.8228N
117.6969W**

TPI = -27.2m (basin floor) | Dist TRc = 1,309m | Composite = 0.990

Strong basin signal on TRc formation margin. Highest composite score in v2. Was v1 #6 - promoted by TPI adjustment. Primary field target.

**#2 - 40.4189N
117.7044W**

TPI = -15.6m (basin floor) | Dist TRc = 3,338m | Composite = 0.980

Basin terrain 3.3km from formation. Second highest v2 score. Consistent terrain signature with Berlin-Ichthyosaur analog.

**#3 - 40.7872N
117.4564W**

TPI = -65.4m (deep basin) | Dist TRc = 13,040m | Composite = 0.953

Deep basin signal but 13km from nearest TRc pixel. Recommend verifying local geology before committing survey time.

Recommended Field Protocol

For each priority target: (1) Verify TRc formation outcrop within 2km using NBMG geologic map. (2) Approach via basin floor - avoid ridge approaches which indicate TPI mismatch. (3) Survey wash exposures and road cuts for bone-bearing horizons in limestone. (4) Record GPS coordinates of any bone fragments and photograph in situ before collection. (5) Submit new localities to PBDB.

GPS / Navigation

Load **paleowave_v2_top50.geojson** into Gaia GPS, Avenza Maps, or Google Earth. Each point includes `composite_score`, `dist_to_triassic_m`, `tpi_15m`, and `tpi_adj`. Filter on `composite_v2 > 0.90` for highest-confidence targets. All coordinates WGS84 decimal degrees.

Phase 3 Recommendations

Formation-stratified model: Separate sub-models for Prida/Favret (40N+) and Luning/Gabbs (38-40N) would address 6 LOO misses concentrated in Luning. Current single-model LOO recall on Prida/Favret core is 100%.

Lithology-matched background: Sample background from carbonate terrain only, not random Nevada pixels. Would reduce AUC inflation and better represent the null hypothesis of carbonate terrain without ichthyosaurs.

Expanded training data: Field-verified new localities from Phase 1-2 field seasons should be submitted to PBDB and used to retrain Phase 3.